

The Architect's Mind in Orbit

How space-station design shapes the crew's mind — and how we measure it

A classroom workshop · Orbital Architecture · KTH Royal Institute of Technology & ESERO Sweden

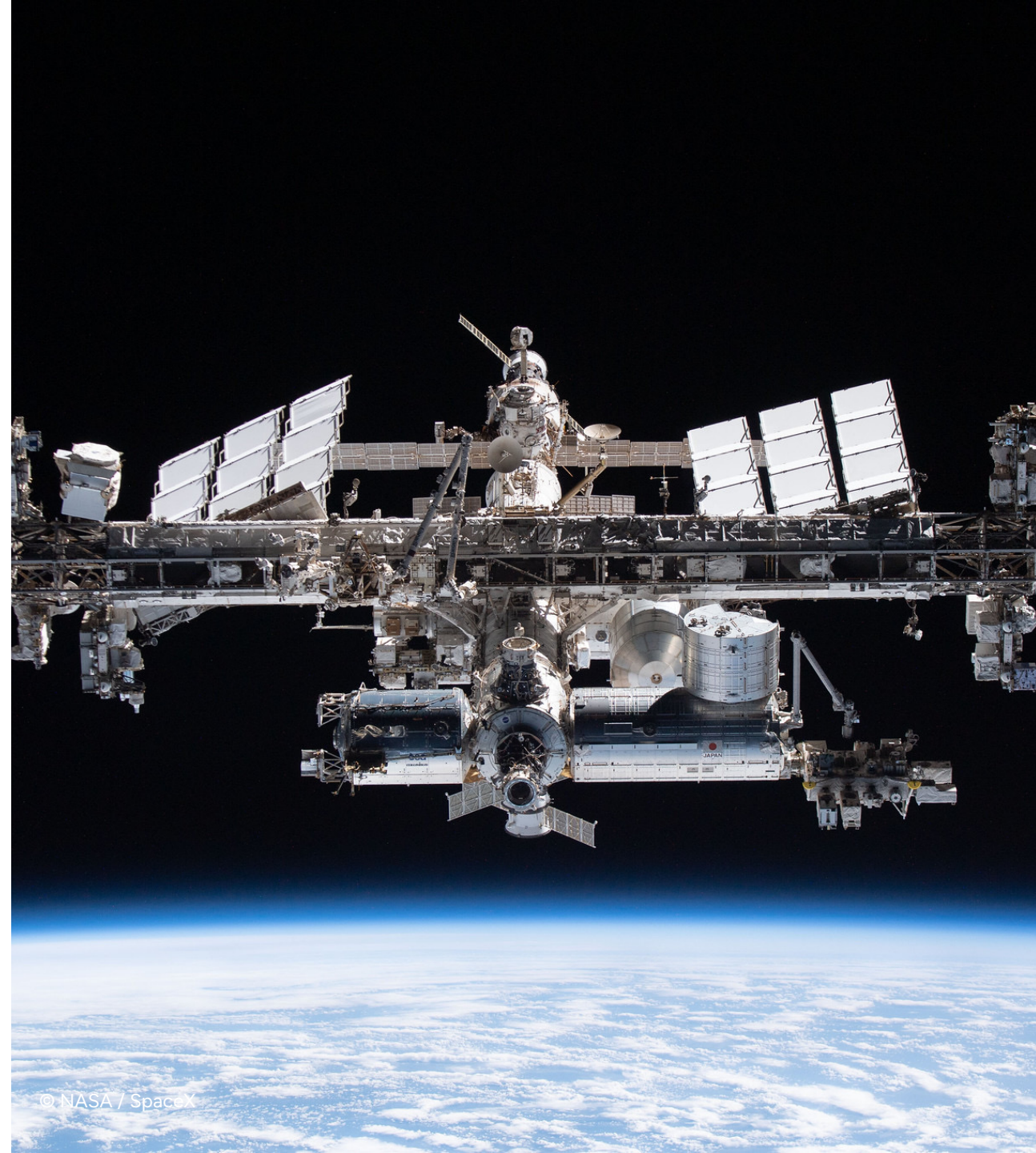
Sweden



© ESA – M. Wandt

What is a space station?

- A spacecraft that stays in orbit so crews can **live and work** there for weeks to months
- A home **and** a workplace — the crew lives with the design every hour of the day
- First station crew (Salyut 1, 1971): 24 days · ISS crews today: about 6 months



A short history of space stations

Salyut

1971

USSR

Mir

1986

USSR / Russia

Tiangong

2021

China

Skylab

1973

USA

ISS

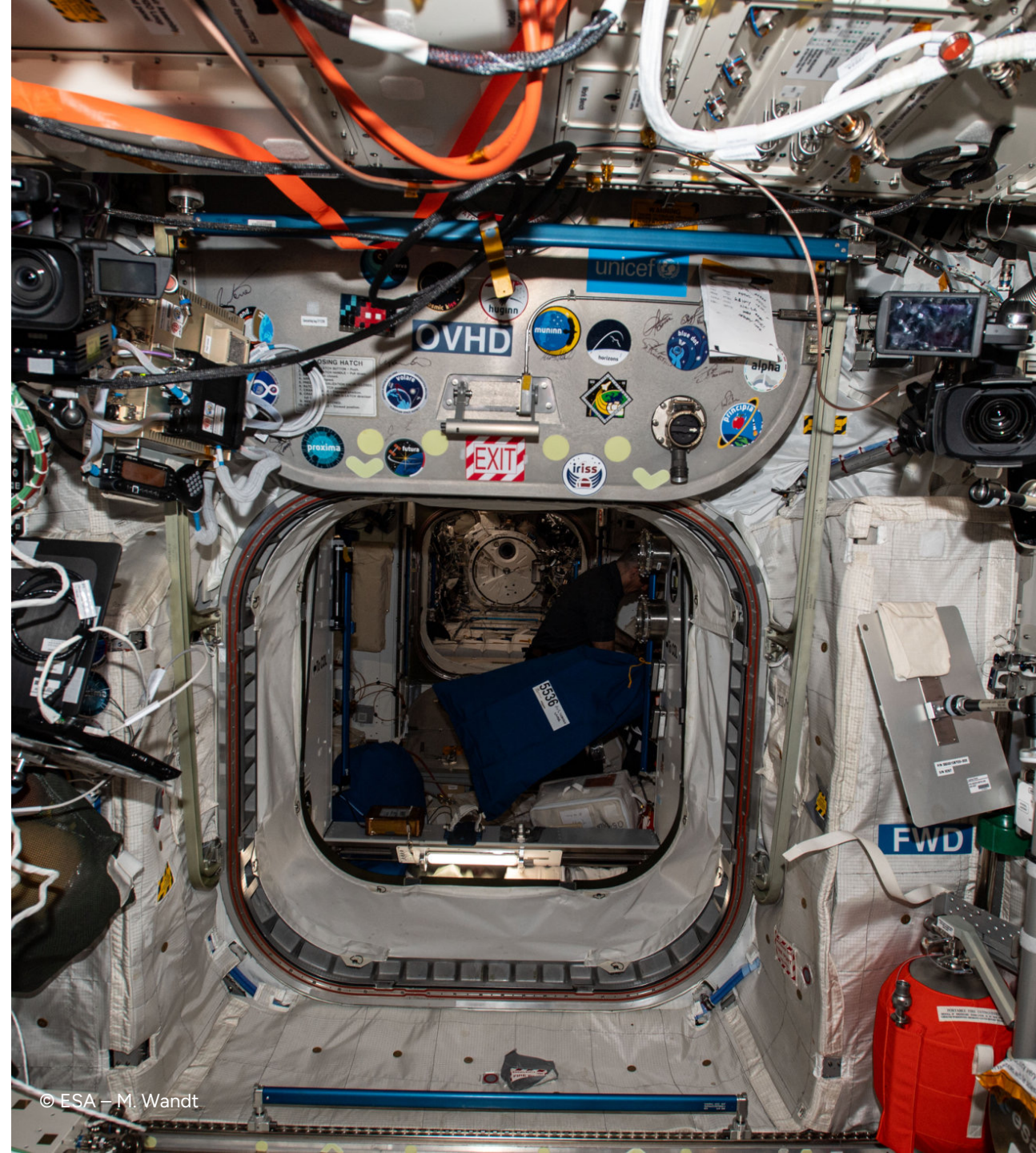
1998

international

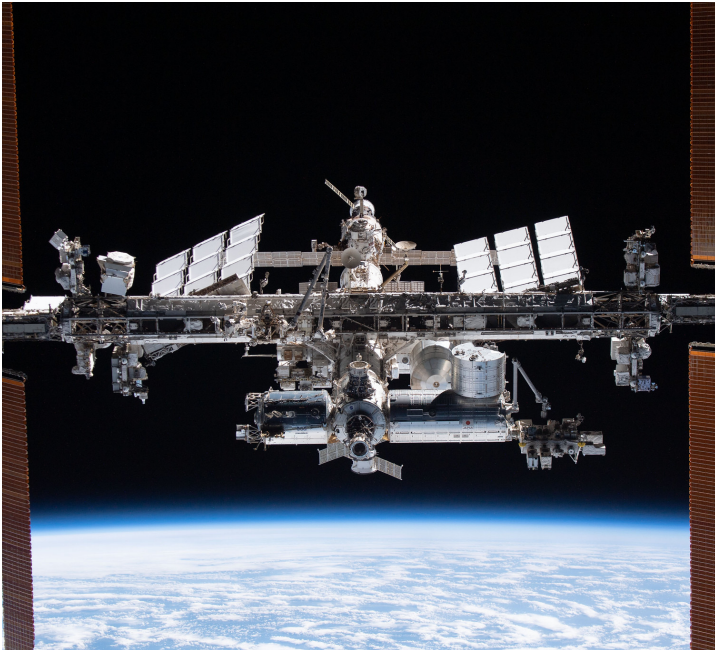
Ask the class where each one was built before revealing the labels.

Machine-centric design

- Cables everywhere
- Cramped and loud
- Complex to operate
- Utilitarian interiors — equipment first, people adapt



From the ground to orbit



The ISS — the international laboratory where the tools you will use today have flown



ESA astronaut Marcus Wandt ran these tests in orbit on the Muninn mission (2024)

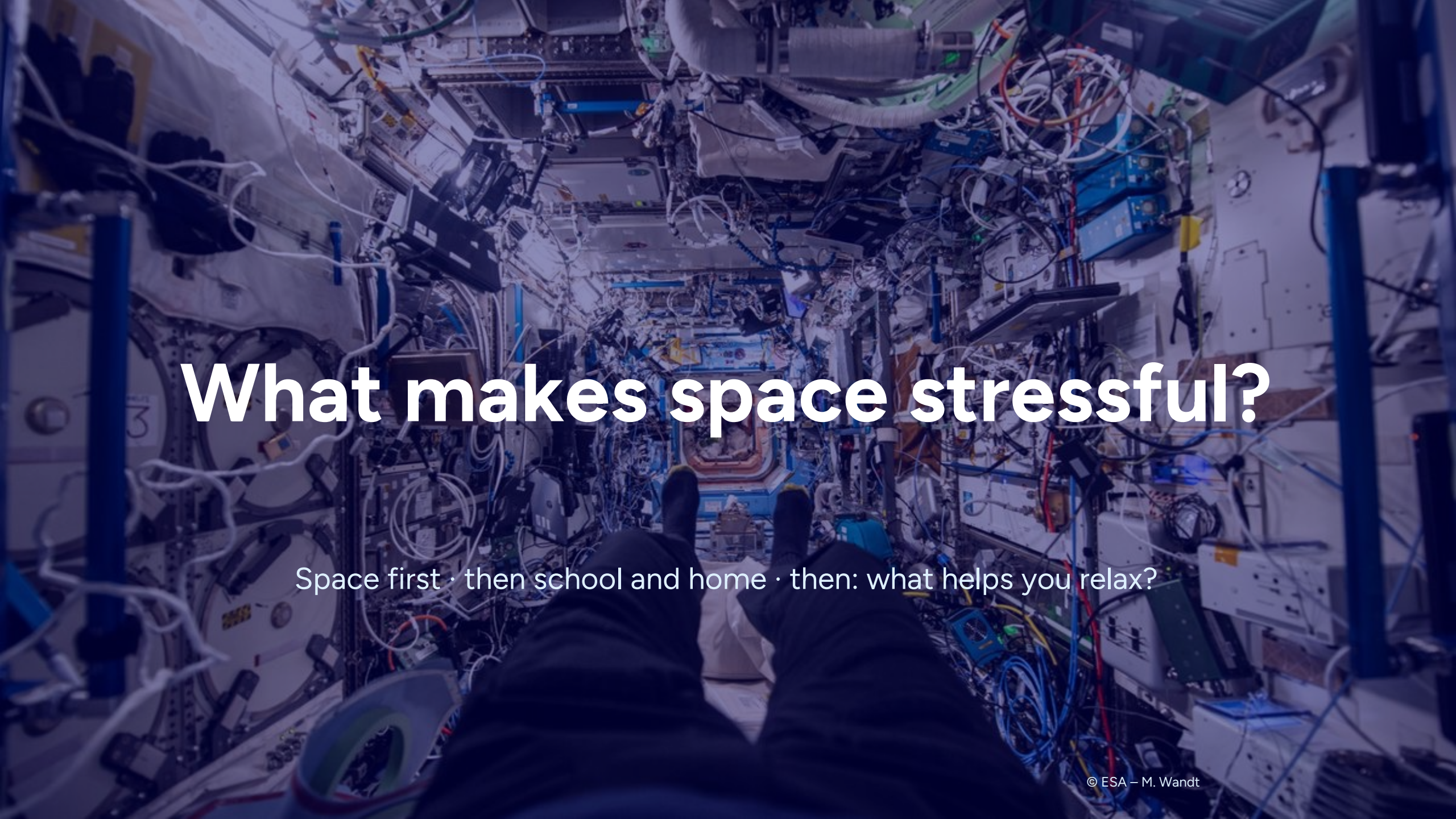


Mars analogue missions — the same tests run in the Utah desert with ISAE-SUPAERO crews

The state of the art today



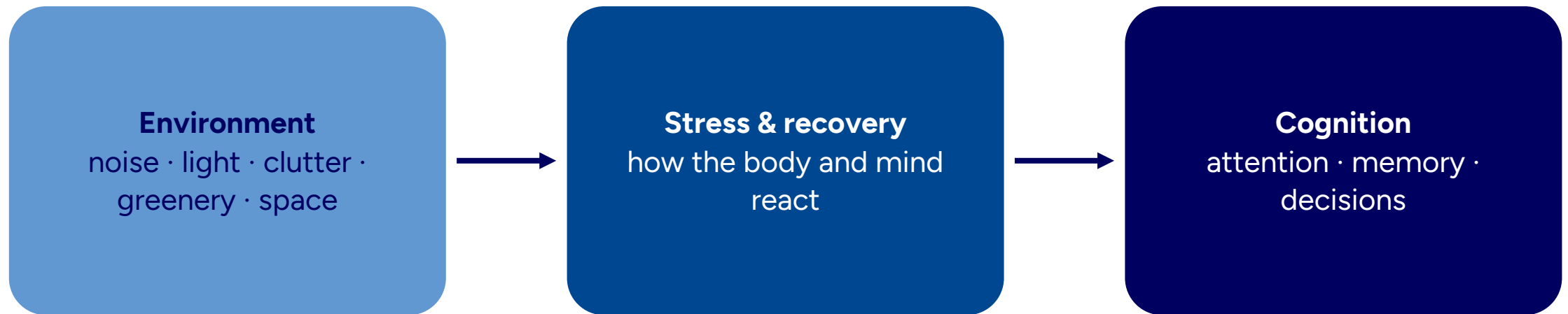
© ESA – M. Wandt · ISS interior photographed by Marcus Wandt in the Destiny laboratory — cables everywhere



What makes space stressful?

Space first · then school and home · then: what helps you relax?

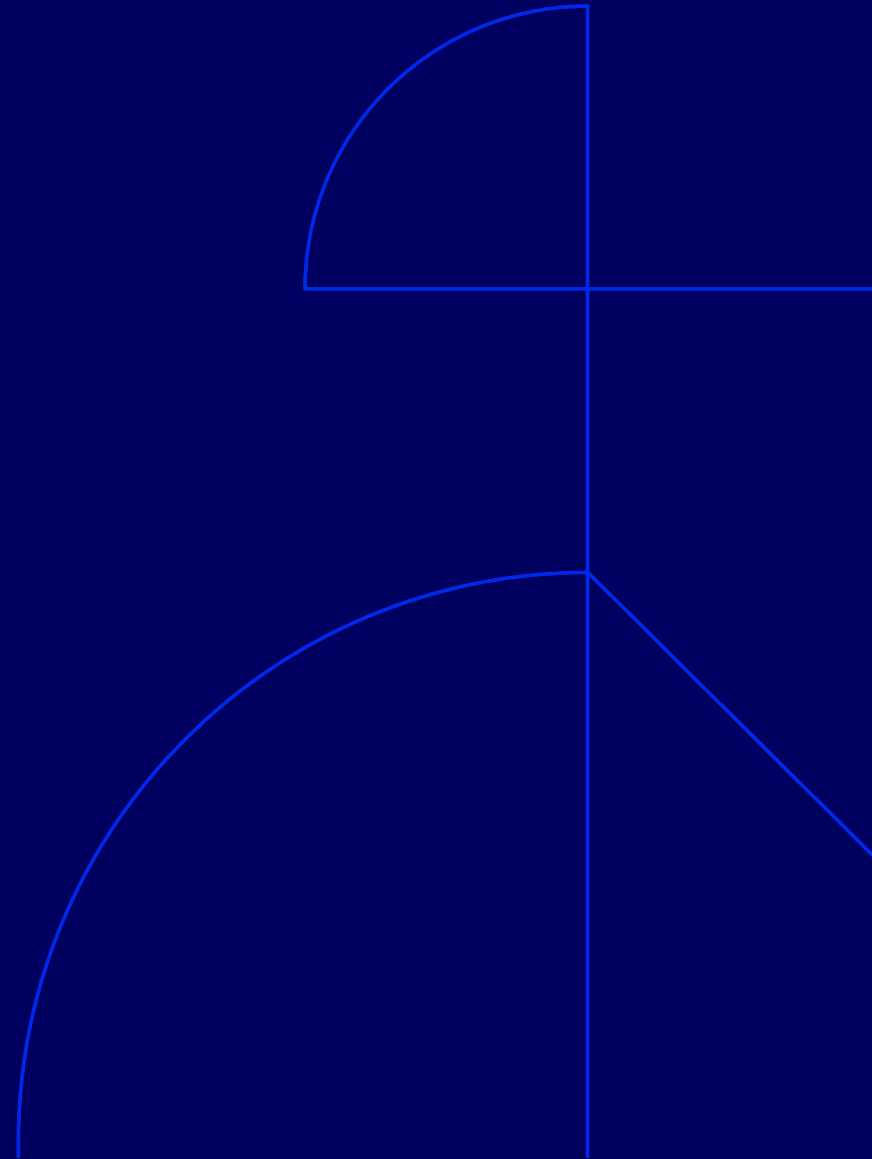
The cognitive link



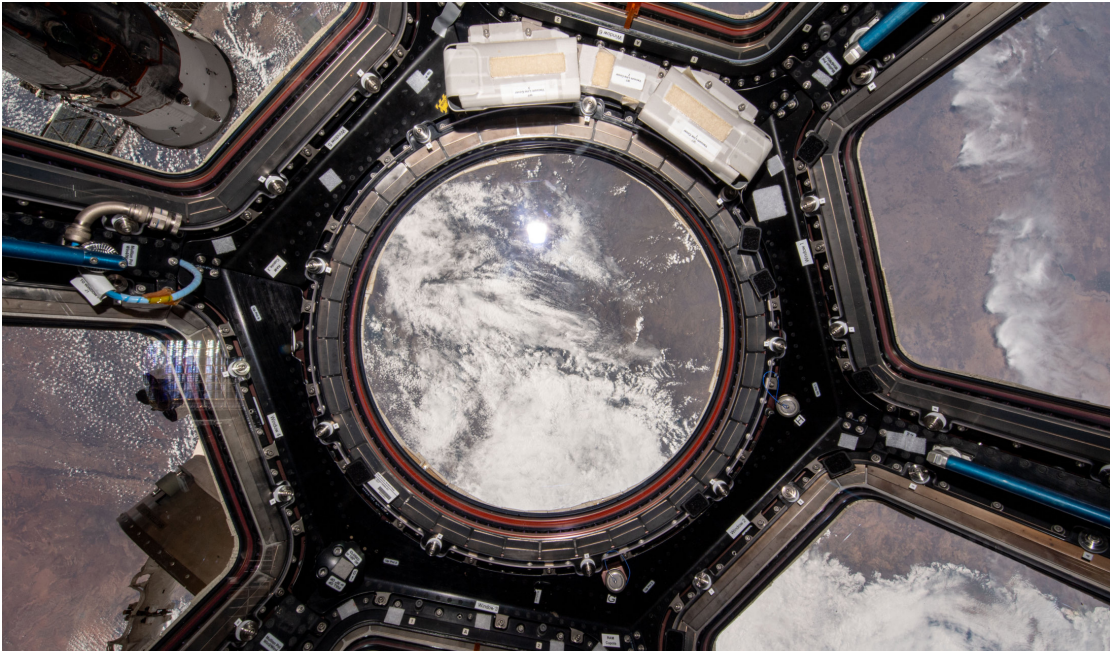
Examples from the board: a noisy cabin raises stress and lowers attention; a view of Earth, plants and good light help you recover.

Better habitats need evidence, not guesswork

Opinions differ — measurements settle it. That is why we test.



What crews live in today



The Cupola — the only window with a real view of Earth



Crew sleeping quarters — a phone-booth-sized private space

The future vision



© NASA – Rick Guidice, NASA Ames Research Center · Rotating habitats with greenery and windows — concept art from the 1970s

Small steps today

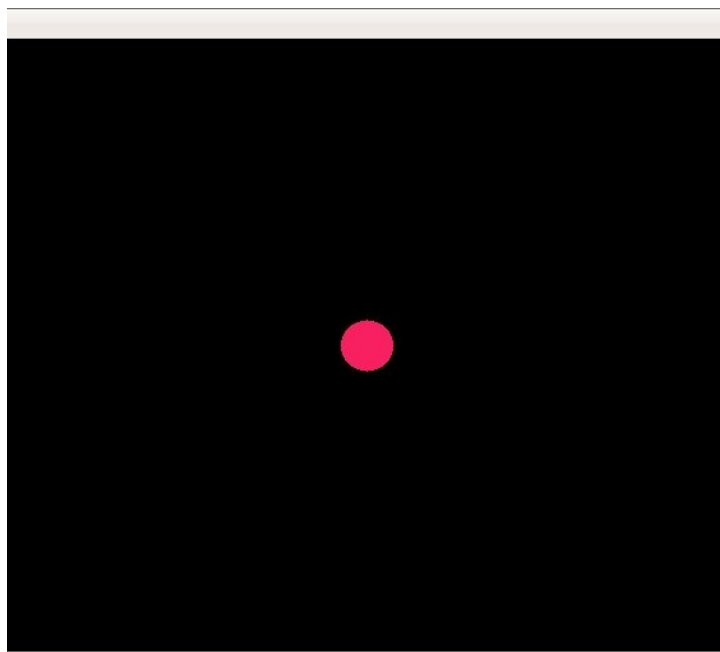
Small quantitative experiments
measure one thing at a time: noise,
light, clutter...



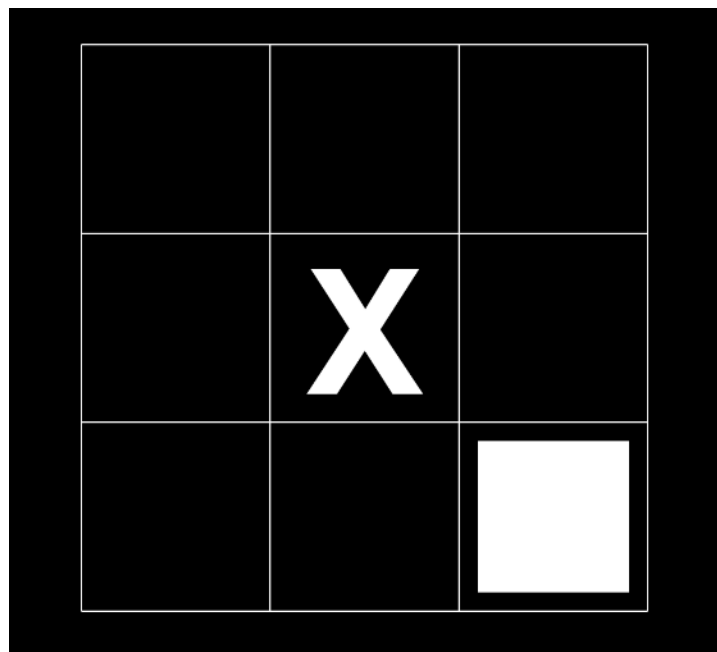
Qualitative change over time
better-designed stations, happier and
sharper crews

We are taking those steps now — and today, so are you.

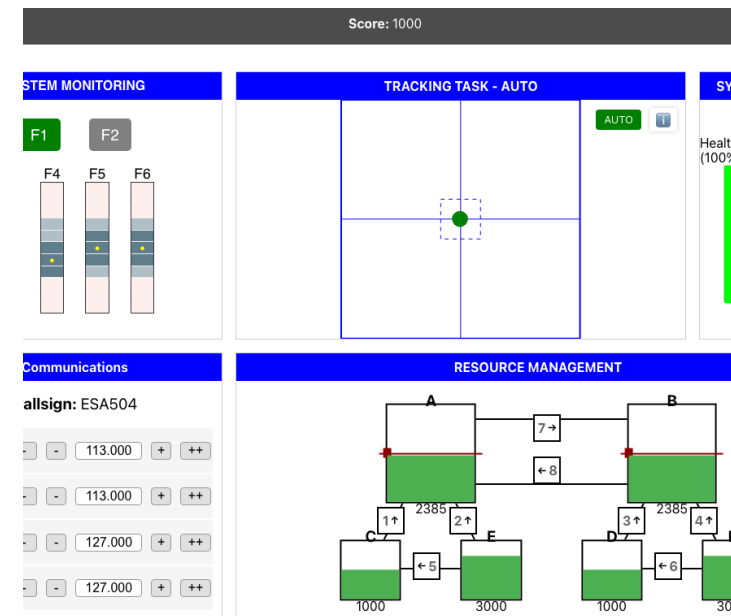
How we test cognition



Reaction Time (PVT-style) — alertness · everyone, as a warm-up



N-Back — working memory · ages 16–18



MATB — executive function · ages 14–15
focus

“How does your math teacher know if you're good at math?” → tests.

Where this research runs



MDRS, Utah — Mars Desert Research Station with ISAE-SUPAERO crews (MDRS 275, 293, 311, 330)



The ISS — Marcus Wandt, ESA Muninn mission, 2024

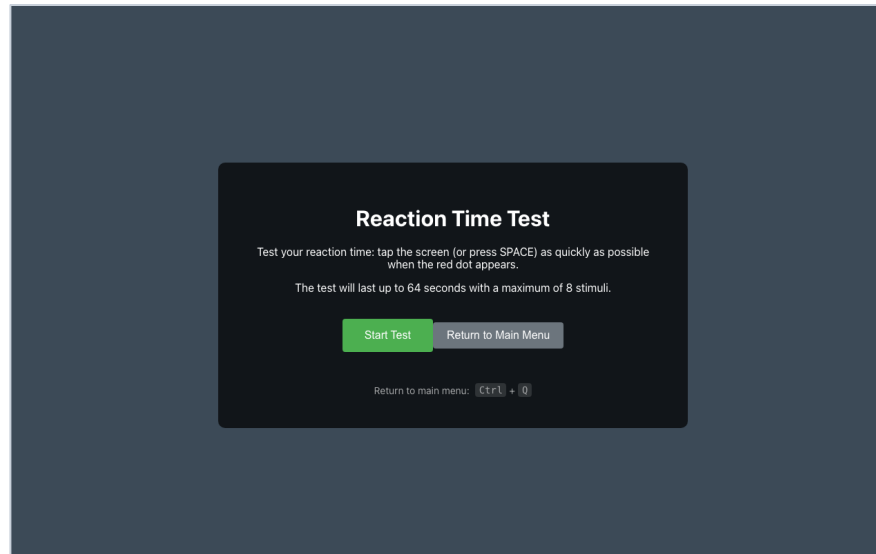
Reaction Time — one round

Tap the screen (or press SPACE) as fast as you can when the red dot appears

One round · 8 dots · about a minute

Everyone is different — confounders

You vs. you: the Condition Lab later compares you with yourself, as a % of your own score

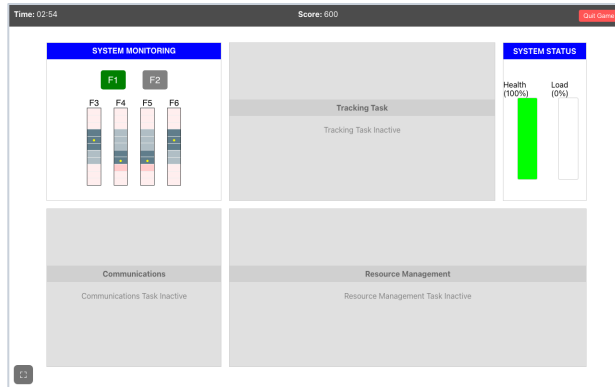


Scan to open
/reaction-default

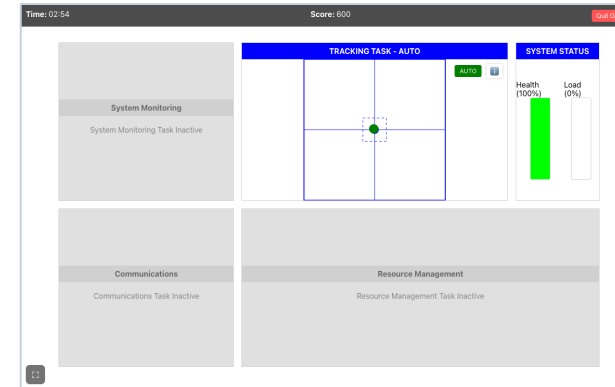
magkosm.github.io/orbarch-edu/reaction-default?lng=en

Start screen. Write your average and fastest time on the handout.

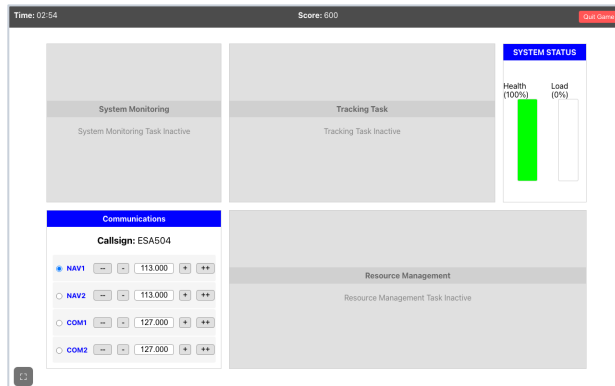
MATB — four tasks at once (teacher demo)



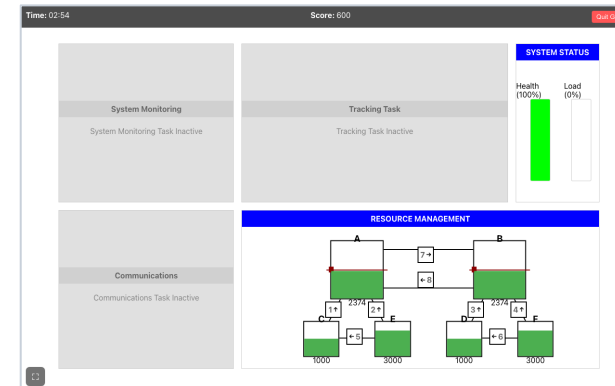
System Monitoring — tap the light or gauge that goes wrong



Tracking — keep the cursor in the square when it says MANUAL



Communications — answer only calls for your own callsign



Resource Management — keep tanks A and B at their target level

Presets → Training modes: one task at a time on the projector. No student QR yet.

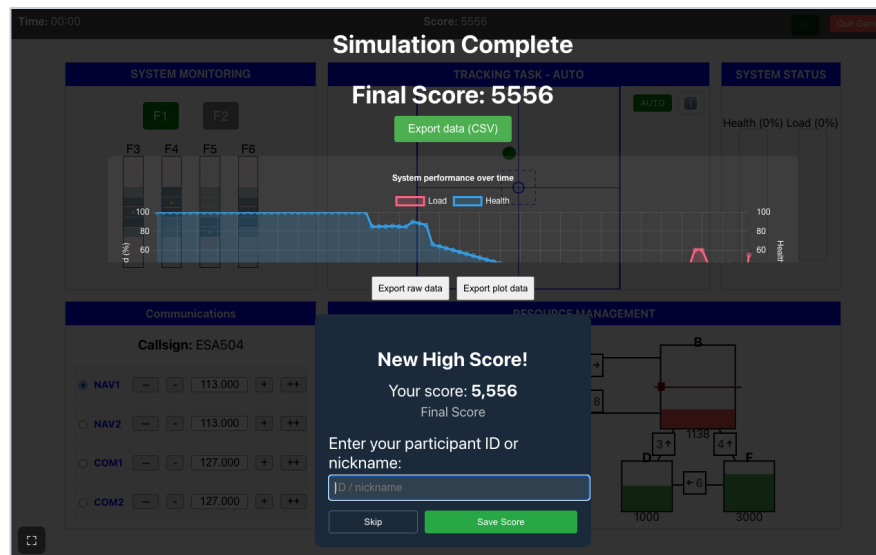
MATB dry run

Everyone: one practice run (2 minutes)

Don't quit — let the time run out

At the end: read the **Final Score** at the top

If a box asks for a name: tap **Skip** — no real names



Scan to open
/2min

magkosm.github.io/orbarch-edu/2min?lng=en

End screen: Final Score at the top; the high-score box can be skipped.

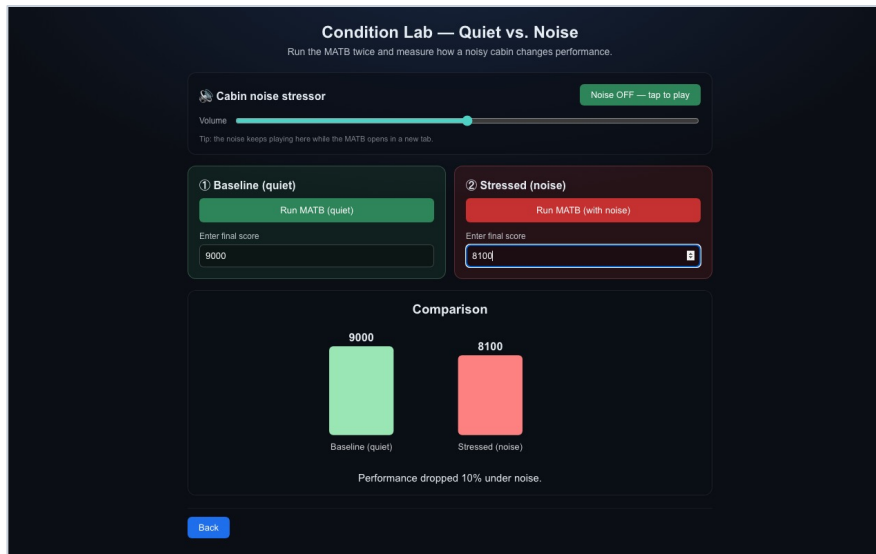
Condition Lab — quiet vs. noise

① Quiet run — all together, speakers off

② Cabin-noise run — all together, speakers on

Type both scores into the Condition Lab

Read the **% change** and copy it to the handout



The screenshot shows the 'Condition Lab — Quiet vs. Noise' interface. At the top, it says 'Run the MATB twice and measure how a noisy cabin changes performance.' Below this is a 'Cabin noise stressor' section with a volume slider and a 'Noise OFF — tap to play' button. A tip states: 'Tip: the noise keeps playing here while the MATB opens in a new tab.' The interface is divided into two main sections: '① Baseline (quiet)' and '② Stressed (noise)'. The 'Baseline (quiet)' section has a green 'Run MATB (quiet)' button and an 'Enter final score' field containing '9000'. The 'Stressed (noise)' section has a red 'Run MATB (with noise)' button and an 'Enter final score' field containing '8100'. Below these is a 'Comparison' section with two colored boxes: a green box for 'Baseline (quiet)' with the value '9000' and a red box for 'Stressed (noise)' with the value '8100'. A text label below the boxes states 'Performance dropped 10% under noise.' At the bottom left is a 'Back' button.

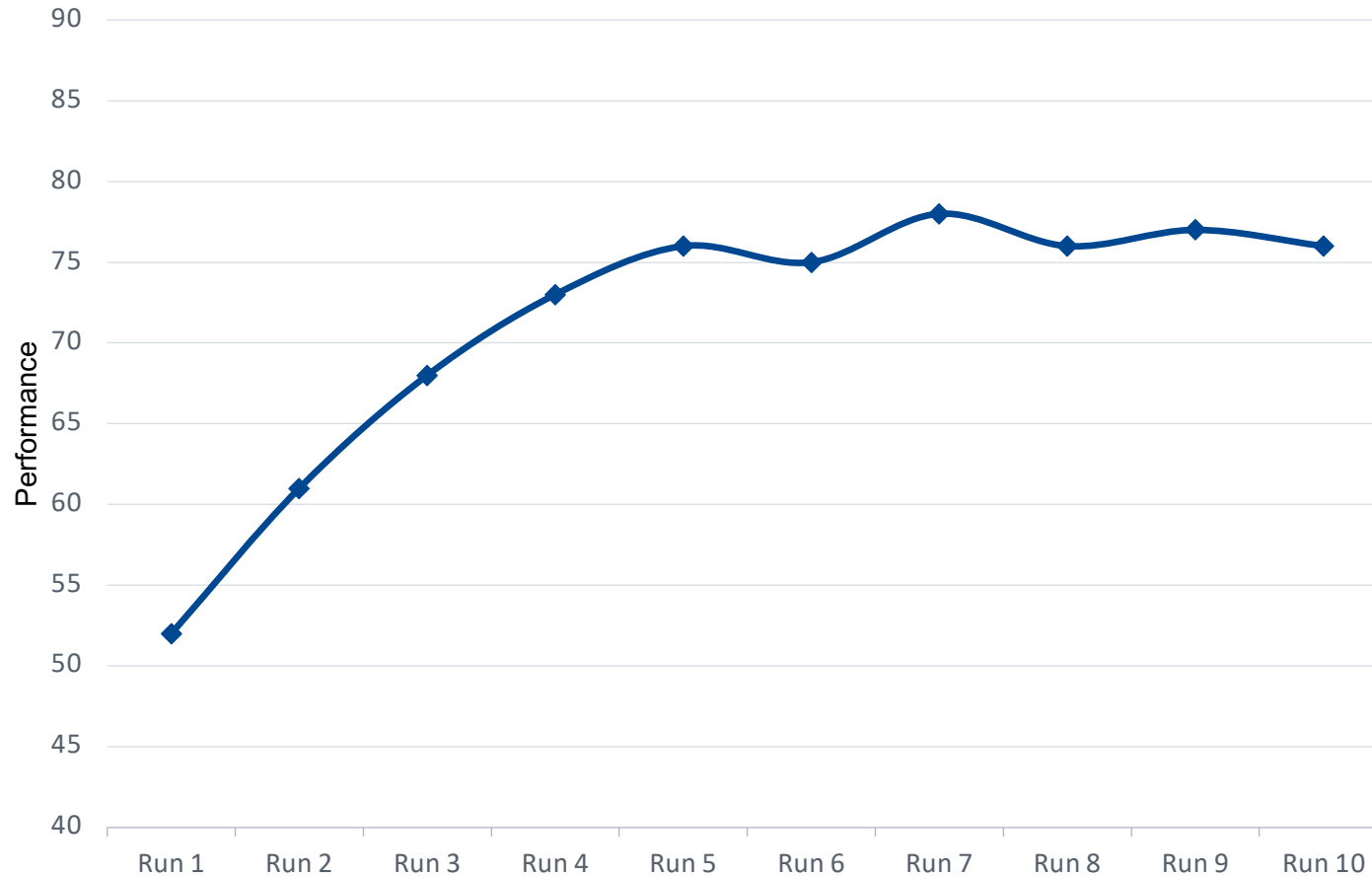


Scan to open
[/condition-lab](https://magkosm.github.io/orbarch-edu/condition-lab?lng=en)

magkosm.github.io/orbarch-edu/condition-lab?lng=en

Both scores entered: the comparison and the % change appear.

Learning and plateau



Like a new sport: you improve fast, then you reach a **plateau**

Daily swings remain — sleep, mood, food

Real studies repeat until scores are stable, **then** test conditions

Today is a taste of that protocol

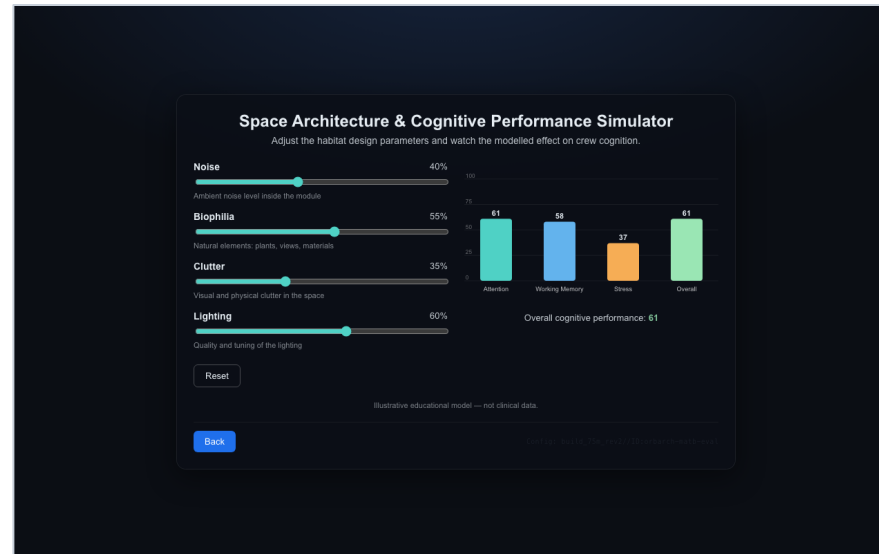
Architecture Simulator

An illustrative model inspired by research

Sliders: noise · light · clutter · biophilia

Watch attention, working memory and stress move

Find the best and the worst cabin



Scan to open
[/simulator](#)

magkosm.github.io/orbarch-edu/simulator?lng=en

Drag the sliders; the bars update live.

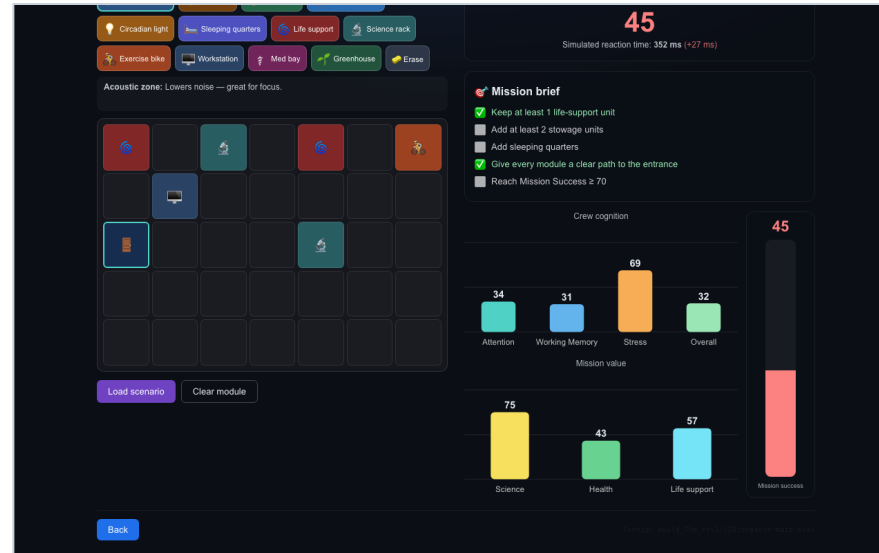
Blueprint Designer — mission brief

Design a station module on the grid

Crew cognition **vs.** science and engineering — every item is a trade-off

Every module needs a free corridor to the **entrance**

Target: **Mission Success ≥ 70**



Scan to open
/blueprint

magkosm.github.io/orbarch-edu/blueprint?lng=en

Load scenario gives a starting layout; the brief checklist turns green as you go.

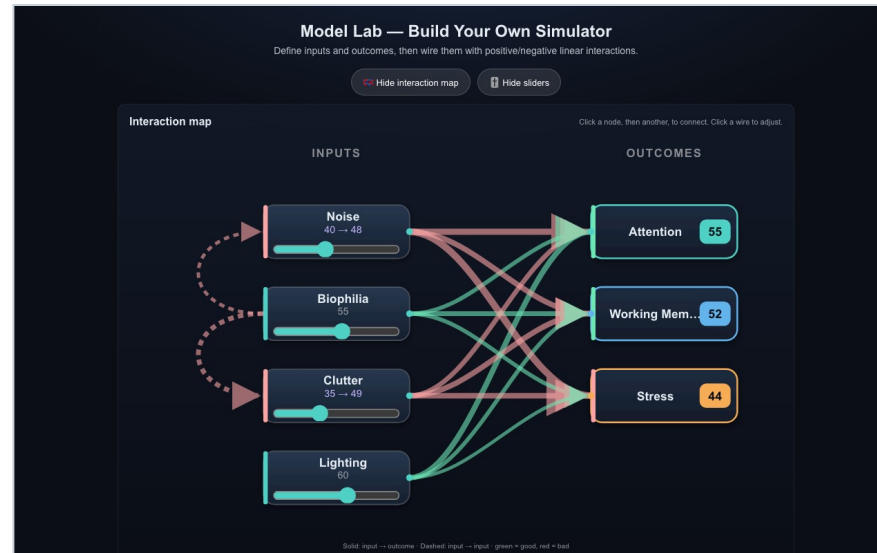
Model Lab (optional)

Build your own simulator

Add inputs and outcomes from the whiteboard list

Green wires help, red wires hurt

If this outcome matters, what should we design for?



Scan to open
[/model-lab](https://magkosm.github.io/orbarch-edu/model-lab?lng=en)

magkosm.github.io/orbarch-edu/model-lab?lng=en

The interaction map shows every link in the model.

Exit ticket

Two sticky notes per student — write your name on each.

① Helps cognition
something in the architecture that
helps the crew's mind

② Hurts cognition
something in the architecture that
hurts it

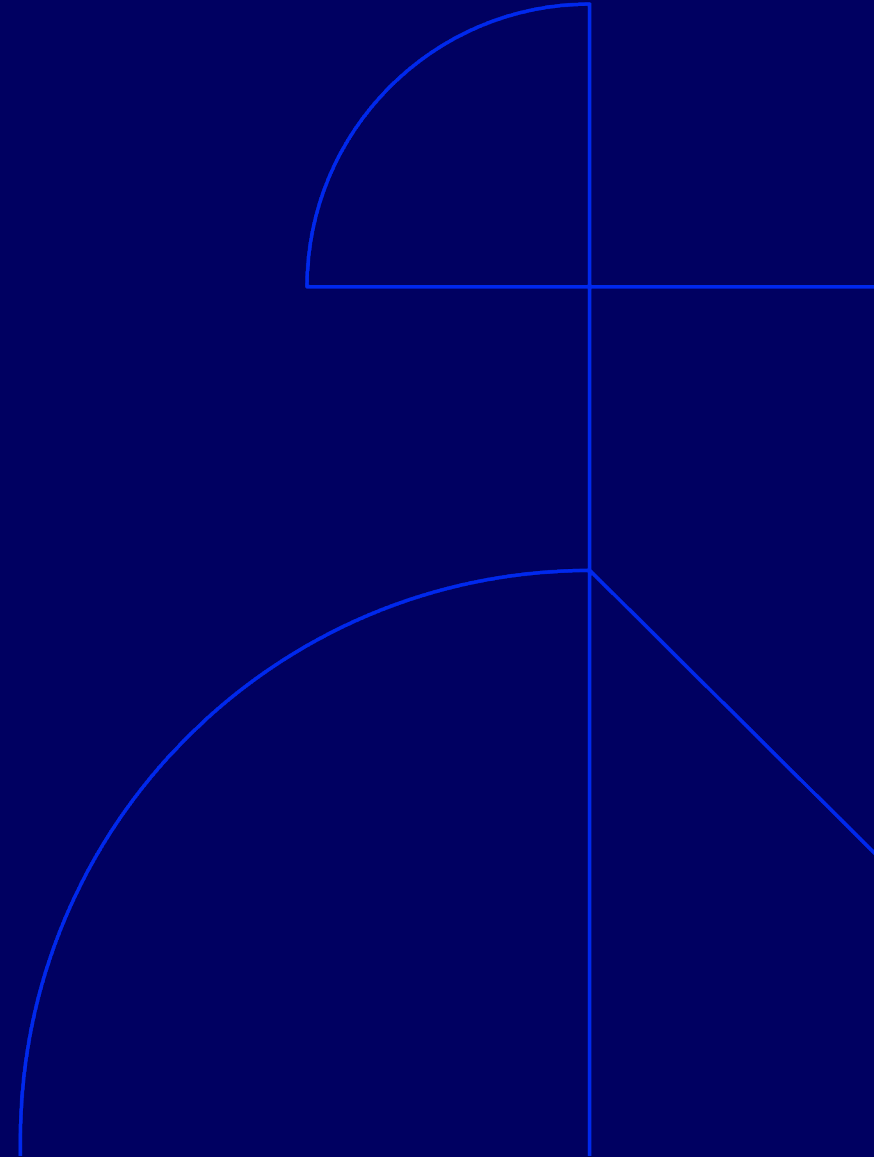
Four things to take home

Space is for everyone.

Architecture is about people.

Design is a series of trade-offs.

You are the future of space exploration.



Thank you

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Open the app at home

